

Rough Bounding

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December 2, 2025

§1 Theory

This handout is about understanding approximate orders of magnitude and avoiding doing work.

Example 1. Prove that for all sufficiently large positive integers n ,

$$\frac{0.0001}{n^{1/6}} < \sum_{x=1}^{\infty} \sum_{y=1}^{\infty} \frac{1}{x^2 + y^3 + n} < \frac{10000}{n^{1/6}}.$$

Solution. The main techniques are to **liberally round up to constant factors**, and to **split into chunks**.

We will split our pairs (x, y) into the following chunks:

1. The set A of pairs with $x^2 + y^3 \leq n$
2. For each nonnegative integer k , the set B_k of pairs with $2^k n < x^2 + y^3 \leq 2^{k+1} n$.

For pairs in A , the summand is, up to a factor of 2, equal to $\frac{1}{n}$. For pairs in B_k , the summand is, up to a factor of 2, equal to $\frac{1}{2^k n}$.

It remains to compute the sizes of A and B_k . For each k , we have

$$\left[\sqrt{2^k n}, 1.1\sqrt{2^k n} \right] \times \left[1, 0.2\sqrt[3]{2^{k+1} n} \right] \subset B_k \subset \left[1, \sqrt{2^{k+1} n} \right] \times \left[1, \sqrt[3]{2^{k+1} n} \right],$$

so $0.01(2^k n)^{5/6} < |B_k| < 100(2^k n)^{5/6}$ for large enough n . Similarly, $0.01n^{5/6} < |A| < 100n^{5/6}$.

We return to the original summation. Terms in A contribute between $0.001/n^{1/6}$ and $100/n^{1/6}$, and terms in B_k contribute between $0.001/(2^k n)^{1/6}$ and $100/(2^k n)^{1/6}$. The sum of $1/(2^k)^{1/6}$ over all k converges to, say, at most 10 since $2^{1/6} > 1.1$. Thus, the total sum is between $0.001/n^{1/6}$ and $1100/n^{1/6}$, as desired. $\square$

Example 2. Prove that if $a_1, a_2, \dots, a_k$ are real numbers summing to zero, then $\sum_{n=1}^{\infty} a_n/n$ is finite, where indices are taken modulo k .

Solution. This time, we split into contiguous length k chunks:

$$\sum_{n=1}^{\infty} \frac{a_n}{n} = \sum_{n=1}^{\infty} \left(\frac{a_1}{nk+1} + \frac{a_2}{nk+2} + \dots + \frac{a_k}{nk+k} \right).$$

The idea is that the coefficients of the $a_1, \dots, a_k$ in each chunk have similar size, so they should cancel

out. We express this idea very directly, as follows:

$$\left| \left(\sum_{i=1}^k \frac{a_i}{nk+1} \right) - \left(\sum_{i=1}^k \frac{a_i}{nk+i} \right) \right| \leq \sum_{i=1}^k a_i \left(\frac{1}{nk+1} - \frac{1}{nk+i} \right) < \frac{1}{n^2} \sum_{i=1}^k |a_i|$$

One thinks about this as follows: we **subtract off the main term** of $a_i/(nk+1)$ from $a_i/(nk+i)$, and then we **analyze the error term**. Anyways, since $\sum_{i=1}^k \frac{a_i}{nk+1} = 0$, the n th chunk of our original summation is at most $\frac{1}{n^2} \sum_{i=1}^k |a_i|$ in magnitude, so the sum is at most $\sum_{n=1}^{\infty} \frac{1}{n^2} \sum_{i=1}^k |a_i|$, which is finite. $\square$

If you get bored of the problems, you can read the appendix where a similar idea is used to derive Stirling's approximation, or you can try to prove the following stronger statement:

Remark. If $a_1, a_2, \dots$ is a sequence of real numbers with $|a_1 + a_2 + \dots + a_n| < n^{0.99}$ for all positive integers n , then $\sum_{n=1}^{\infty} a_n/n$ is finite.

The solution to this remark demonstrates the power of **nory fhzzngvba** (rot13), which you should think of as a quicker way of accessing the cancellation in the above solution.

§2 Problems

When I ask you estimate a function $f(n)$, I am requesting an explicit function g such that for some constant C , one has $g(n)/C \leq f(n) \leq Cg(n)$.

Problem 2.1. Consider the set of k -term arithmetic progressions with all terms in $\{1, 2, \dots, n\}$. Prove that for some constant C_k independent of n , at most $C_k n^2$ pairs of such arithmetic progressions have at least 2 common terms.

Problem 2.2. Suppose that $\left| \log n - \sum_{p \leq n} \frac{\log p}{p} \right| < 100$ for all n . Estimate the number of primes below n .

Problem 2.3. Estimate the number of triples (a, b, c) of positive integers with $abc \leq n$.

Problem 2.4. Estimate $\sum_{x=1}^{\infty} \frac{1}{|x^2 - n^2 - n|}$.

Problem 2.5. Let n and k be positive integers. Prove that for $a_1, \dots, a_n \in [1, 2^k]$ one has

$$\sum_{i=1}^n \frac{a_i}{\sqrt{a_1^2 + \dots + a_i^2}} \leq 4\sqrt{kn}.$$

§3 Additional Practice

Problem 3.1. Estimate the largest positive integer t for which $\binom{t}{n} 2^{-n^2} < 1$.

Problem 3.2. Estimate the number of pairs (a, b) of positive integers with $a^b \leq n$.

Problem 3.3. Prove that $\int_0^{\infty} (x^{-s} - [x]^{-s}) dx$ converges when $s > 0$.

Problem 3.4. Let $f(x)$ be an infinitely differentiable function with period 1. Prove that for all positive integers k , there exists a constant C_k such that

$$\left| \int_0^1 f(x) \sin(\pi n x) dx \right| \leq C_k n^{-k}.$$

Problem 3.5. For a real number a , let $F_a(x) = \sum_{n \geq 1} n^a e^{2^n x^{n^2}}$ for $0 \leq x < 1$. Find a real number c such that

$$\begin{aligned} \lim_{x \rightarrow 1^-} F_a(x) e^{-1/(1-x)} &= 0 \quad \text{for all } a < c, \text{ and} \\ \lim_{x \rightarrow 1^-} F_a(x) e^{-1/(1-x)} &= \infty \quad \text{for all } a > c. \end{aligned}$$

§A Stirling's Approximation

Example 3. Prove that $n!$ is estimated by $\frac{n^n}{e^n} \sqrt{n}$.

Solution. We begin with the following integral approximation:

$$\log n! = \log 1 + \log 2 + \cdots + \log n \approx \int_1^n \log x dx.$$

The integral has length about n and is usually about $\log n$, so it's about $n \log n$. The error to this naive estimate is given by **integration by parts**:

$$\int_1^n \log x dx = n \log n - \int_1^n 1 dx = n \log n - n + 1.$$

We now compute the error between the integral and our summation, which is

$$\int_1^n \log x dx - (\log 1 + \log 2 + \cdots + \log n) = \sum_{k=2}^n \int_0^1 \log \left(\frac{k-t}{k} \right) dt.$$

Since $\log(1+\varepsilon) \approx \varepsilon$, with error at most ε^2 for $\varepsilon < 0.5$, the above integral is approximately

$$\sum_{k=2}^n \int_0^1 (-t/k) dt = -\frac{1}{2} (1/2 + 1/3 + \cdots + 1/n)$$

with error bounded by the constant $\sum_{k=2}^{\infty} \int_0^1 (t/k)^2 dt$. Since the harmonic series is within some constant of $\log n$, we conclude that for some $C > 0$,

$$\left| \log n! - \left(n \log n - n + \frac{1}{2} \log n \right) \right| < C,$$

as desired

□